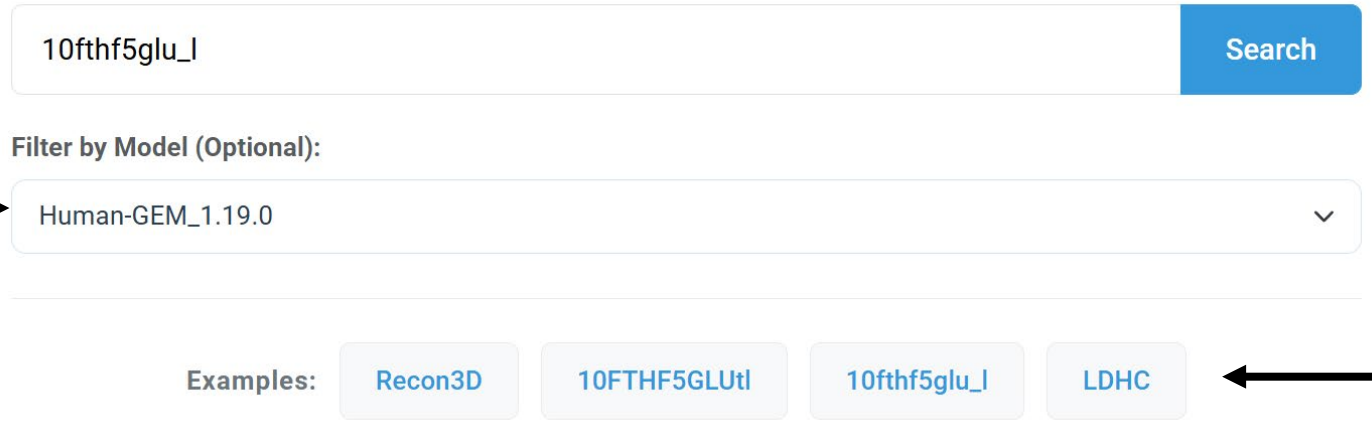
help.html figure

Search box
(e.g. 10fthf5glu_l)

choose model
(e.g. Human-GEM_1.19.0)



Load examples

Search results summary box

Search Results for "10fthf5glu_l"
Found 0 models, 3 reactions, 1 metabolites, 0 genes

Reactions (3)

10FTHF5GLUtl 5-glutamyl-10FTHF transport, lysosomal

Show $\Delta_r G°$

1 models

View $\Delta_r G°$ values

Download results for the GEM model

Reaction classification overview chart

$\Delta_rG°$ distribution analysis chart

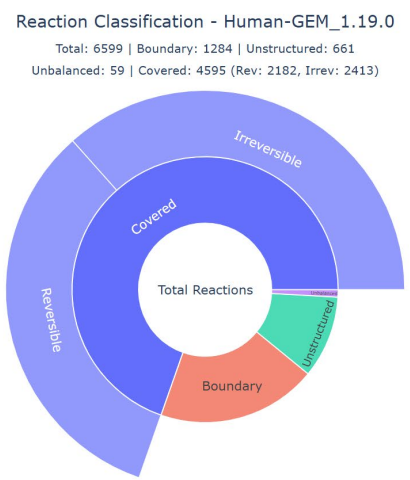
The definition of four class reactions

The compartment conditions use for prediction

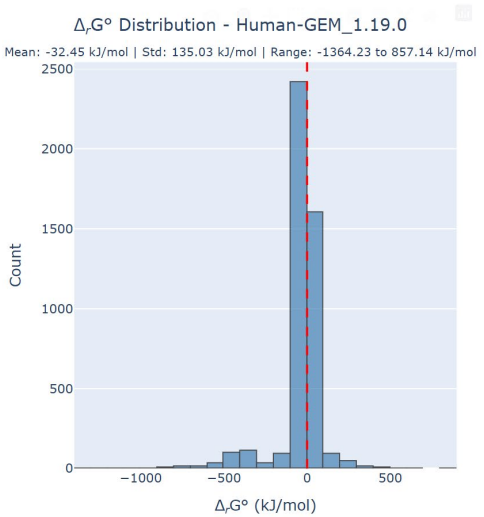
Human-GEM_1.19.0

Download

Reaction Classification Overview



$\Delta_rG°$ Distribution Analysis



Reaction Classification Guide

Boundary reaction
Reaction involving extracellular metabolites (e.g., exchange).

Unstructured reaction
Cannot be checked for balance due to missing formulas or charges.

Unbalanced reaction
Has complete data but is not mass or charge balanced.

Covered reaction
Fully defined and balanced; suitable for further analysis.

Compartment Conditions

Cytosol pH 7.2 T 298K I 0.25M pMg 14 E 0.000V	**Extracellular** pH 7.4 T 298K I 0.25M pMg 14 E 0.030V	**Nucleus** pH 7.2 T 298K I 0.25M pMg 14 E 0.000V	**Endoplasmic Reticulum** pH 7.2 T 298K I 0.25M pMg 14 E 0.000V
Golgi Apparatus pH 6.3 T 298K I 0.25M pMg 14 E 0.000V	**Lysosome** pH 5.5 T 298K I 0.25M pMg 14 E 0.019V	**Mitochondria** pH 8.0 T 298K I 0.25M pMg 14 E -0.155V	**Inner Mitochondria** pH 8.0 T 298K I 0.25M pMg 14 E -0.155V
Peroxisome pH 7.0 T 298K I 0.25M pMg 14 E 0.012V			

Reaction BiGG ID

$\Delta_r G°$ data Table

10FTHF5GLUtl 5-glutamyl-10FTHF transport, lysosomal

Hide $\Delta_r G°$

1 models

Model	Reaction Equation	$\Delta_r G°$ (kJ/mol)	SD (kJ/mol)
Human-GEM_1.19.0	`10fthf5glu_c --> 10fthf5glu_l`	-447.94	0.00

Metabolite BiGG ID

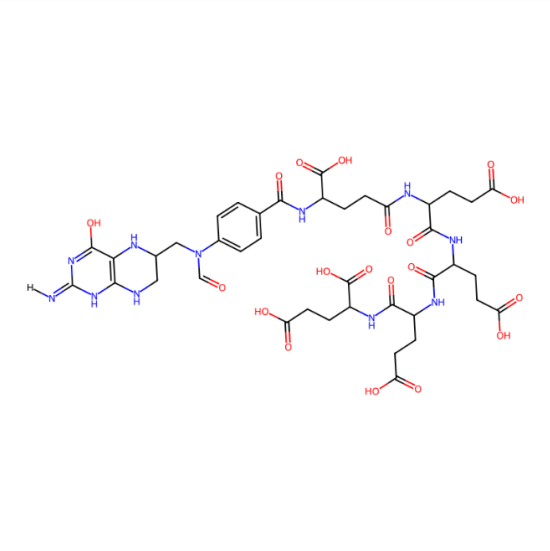
10fthf5glu_l

Name: 10-formyltetrahydrofolate-[Glu](5)

Universal ID: 10fthf5glu

Formula: C40H45N11O19

2D chemical structure image



Related Reactions:

10FTHF5GLUtl	Show $\Delta_r G^\circ$ 1 models
GGH_10FTHF5GLUl	Show $\Delta_r G^\circ$ 1 models
GGH_10FTHF6GLUl	Show $\Delta_r G^\circ$ 1 models

Reactions involving this metabolite

Gene ID

Reaction BiGG ID

GPR information

ENSG00000166796 (LDHC)

Gene ID: ENSG00000166796

Associated Reactions: 4

Gene Name: LDHC

Models: Human-GEM_1.19.0

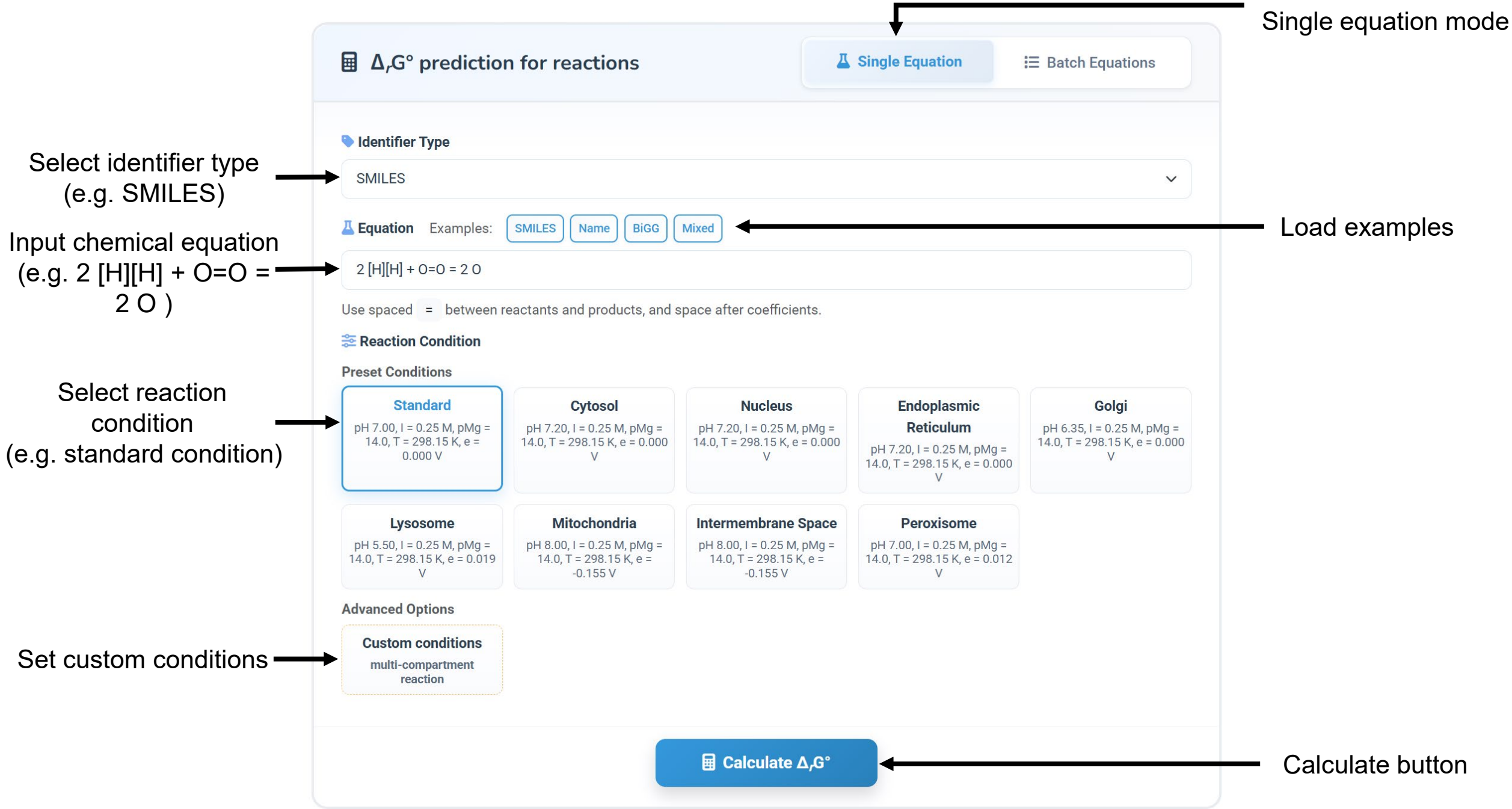
Associated Reactions:

2H3MVc
Formation of 2-Hydroxy-3-Methyl-Valerate
GPR: ENSG00000111716 or ENSG00000134333 or ENSG00000166796 or ENSG00000166800 or ENSG00000166816 or ENSG00000171989 (Human-GEM_1.19.0)

2HIVc
Formation of 2-Hydroxy-Isovalerate
GPR: ENSG00000111716 or ENSG00000134333 or ENSG00000166796 or ENSG00000166800 or ENSG00000166816 or ENSG00000171989 (Human-GEM_1.19.0)

GLXO1
None
GPR: ENSG00000111716 or ENSG00000166796 or ENSG00000166800 or ENSG00000166816 or ENSG00000171989 (Human-GEM_1.19.0)

LDH_L
(S)-Lactate:NAD+ oxidoreductase
GPR: ENSG00000111716 or ENSG00000134333 or ENSG00000166796 or ENSG00000166800 or ENSG00000171989 (Human-GEM_1.19.0)


Single equation mode
ΔrG° prediction for reactions
Single Equation
Batch Equations
Identifier Type
Select identifier type (e.g. SMILES)
SMILES
Equation
Examples:
SMILES
Name
BiGG
Mixed
Load examples
Input chemical equation (e.g. 2 [H][H] + O=O = 2 O)
2 [H][H] + O=O = 2 O
Use spaced = between reactants and products, and space after coefficients.
Reaction Condition
Preset Conditions
Select reaction condition (e.g. standard condition)
Standard
pH 7.00, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.000 V
Cytosol
pH 7.20, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.000 V
Nucleus
pH 7.20, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.000 V
Endoplasmic Reticulum
pH 7.20, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.000 V
Golgi
pH 6.35, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.000 V
Lysosome
pH 5.50, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.019 V
Mitochondria
pH 8.00, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = -0.155 V
Intermembrane Space
pH 8.00, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = -0.155 V
Peroxisome
pH 7.00, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.012 V
Advanced Options
Set custom conditions
Custom conditions
multi-compartment reaction
Calculate ΔrG°
Calculate button

Calculation Result

$\Delta_r G°$

-529.06

kJ/mol

Standard Deviation

3.96

kJ/mol

Identifier: SMILES

Equation: `2 [H][H] + O=O = 2 O`

Condition:

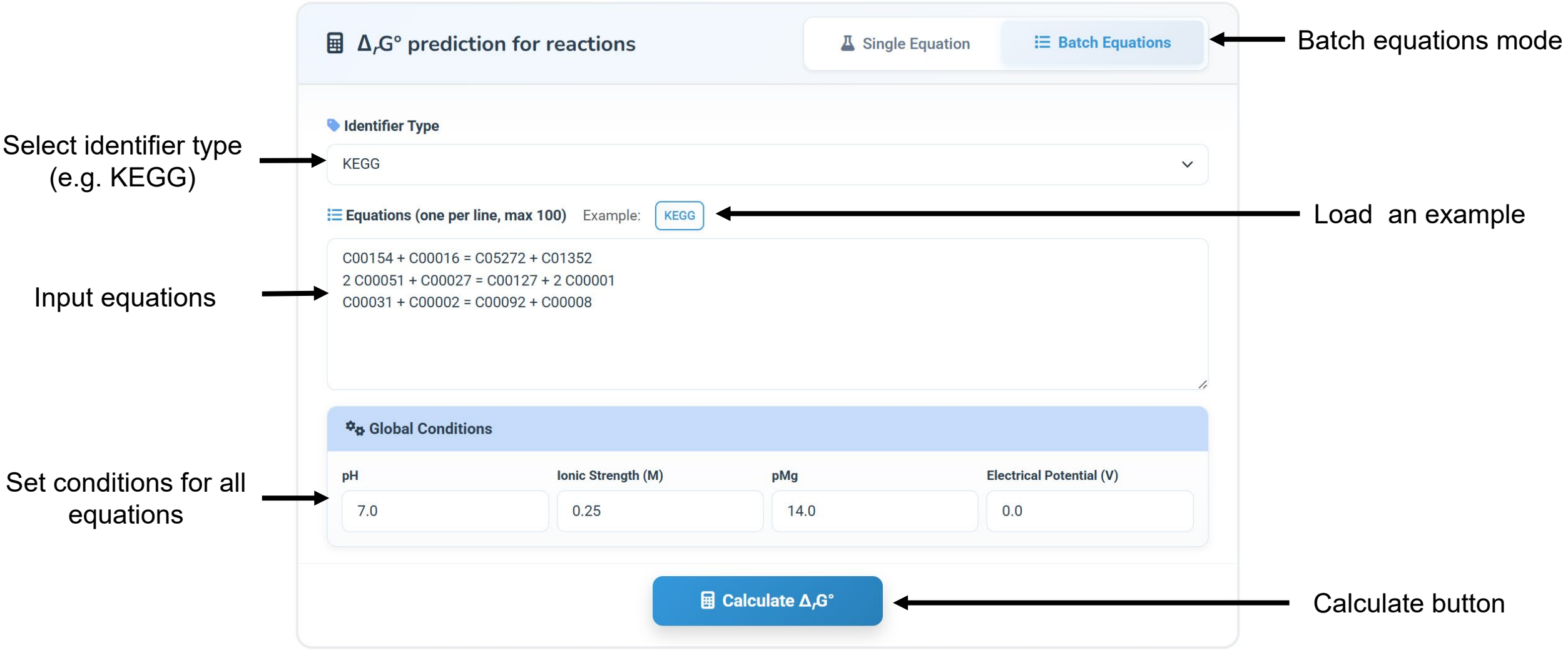
Standard

pH 7.00, I = 0.25 M, pMg = 14, T = 298.15 K, e = 0.000 V

Copy Result

Close

Calculate Again


Δ$_r$G° prediction for reactions
Single Equation
Batch Equations
Batch equations mode
Identifier Type
Select identifier type (e.g. KEGG)
KEGG
Equations (one per line, max 100)
Example:
KEGG
Load an example
C00154 + C00016 = C05272 + C01352
2 C00051 + C00027 = C00127 + 2 C00001
C00031 + C00002 = C00092 + C00008
Input equations
Global Conditions
pH
Ionic Strength (M)
pMg
Electrical Potential (V)
7.0
0.25
14.0
0.0
Set conditions for all equations
Calculate Δ$_r$G°
Calculate button

Batch Results

Equation	$\Delta_r G°$ (kJ/mol)	Std Dev (kJ/mol)
C00154 + C00016 = C05272 + C01352	**40.22**	4.79
2 C00051 + C00027 = C00127 + 2 C00001	**-310.04**	4.54
C00031 + C00002 = C00092 + C00008	**-15.48**	0.36

Identifier: KEGG

Global Conditions: pH = 7.0, I = 0.25 M, pMg = 14.0, T = 298.15 K, e = 0.0 V

3 successful, 0 failed

Download CSV

Close

Calculate Again